



LDMOS RF Power Transistor

1. Description

HTL7G06S006P is a MOSFET type transistor specifically designed for VHF/UHF RF Power Amplifier applications.

This device has an internal monolithic Zener diode from gate to source for ESD protection.

HTL7G06S006P

100-600MHz, 6W, 7.2V
WIDE BAND
RF POWER LDMOS TRANSISTOR

2. Features

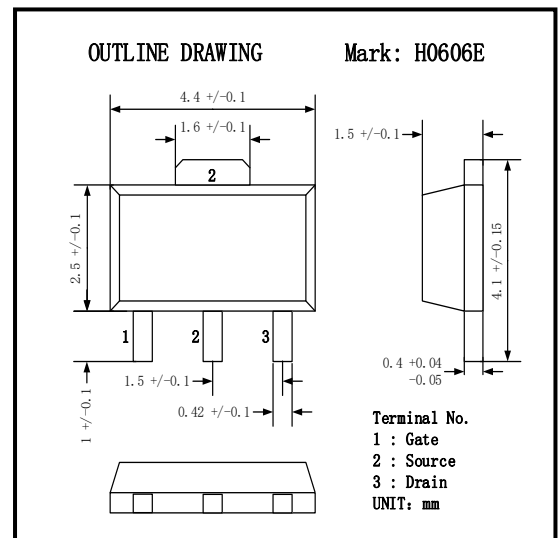
Frequency (MHz)	VDD (V)	Pin (W)	Pout (W)	Eff (%)
175 ⁽¹⁾	7.4	0.32	6.5	66
520 ⁽²⁾	7.4	0.32	5.5	61

1. Measured in 175MHz narrowband reference circuit.

2. Measured in 520MHz narrowband reference circuit.

3. Application

- For output stage of high power amplifiers in VHF/UHF Band mobile radio sets.
- For drive stage of high power amplifiers in Universal Broadband.



4. Absolute Maximum Ratings

PARAMETER	SYMBOL	CONDITIONS	RATINGS	UNIT
Drain to source voltage	V_{DSS}	$V_{gs} = 0V$	20	V
Gate to source voltage	V_{GSS}	$V_{ds} = 0V$	-5 ~ 10	V
Operating Voltage	V_{DD}	-	9	V
Storage temperature	T_{stg}	-	-55 ~ 150	°C
Operating Junction Temperature	T_J	-	-40 ~ 150	°C
Thermal resistance	$Z_{th(j-c)}$	Junction to case	6.5	°C/W

5. Electrical Characteristics

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT
Breakdown Voltage	$V_{(BR)DDS}$	$V_{GG} = 0V, I_D = 39.6\mu A$	20	-	-	V
Zero Gate Voltage Drain Leakage Current	I_{DSS}	$V_{DD} = 12V, V_{GG} = 0V$	1	-	-	μA
Gate-Source Leakage Current	I_{GSS}	$V_{DD} = 0V, V_{GG} = 10V$	1	-	-	μA
Gate Threshold Voltage	V_{th}	$V_{DD} = V_{GG}, I_D = 39.6\mu A$	0.8	1.55	2.6	V
Output Power	P_{out}	$V_{DD} = 7.4V, P_{in} = 0.31W$ $f = 175MHz, I_{DQ} = 500mA$	-	6.5	-	W
Drain Efficiency	η_d		-	66	-	%
Output Power	P_{out}	$V_{DD} = 7.4V, P_{in} = 0.31W$ $f = 520MHz, I_{DQ} = 500mA$	-	5.5	-	W
Drain Efficiency	η_d		-	61	-	%

6. ESD Protection Characteristics

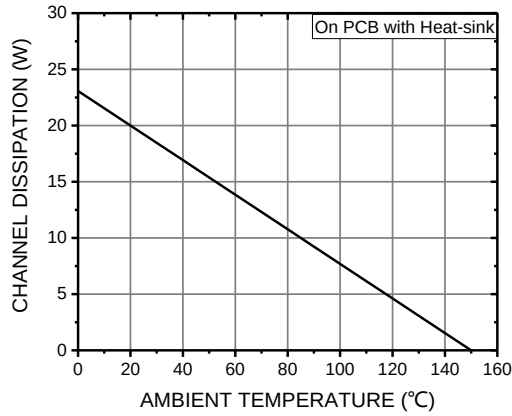
Test Methodology	Class
Human Body Model (per JESD22-A114)	1B
Machine Model (per EIA/JESD22-A115)	A
Charge Device Model (per JESD22-C101)	III

7. Load Mismatch (*in Huatai Test Fixture*)

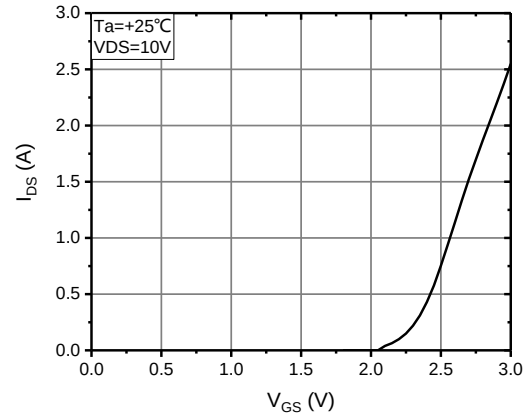
Test Methodology	Results
VSWR = 10:1 at all Phase Angles CW: VDD=8.4Vdc, IDQ=500mA, f=175MHz, Pout =37dBm.	No Device Degradation
VSWR = 10:1 at all Phase Angles CW: VDD=8.4Vdc, IDQ=500mA, f=520MHz, Pout =37dBm.	No Device Degradation

8. Typical Characteristics

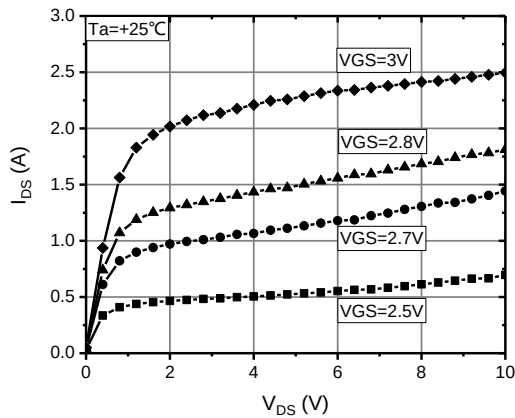
CHANNEL DISSIPATION VS.
AMBIENT TEMPERATURE



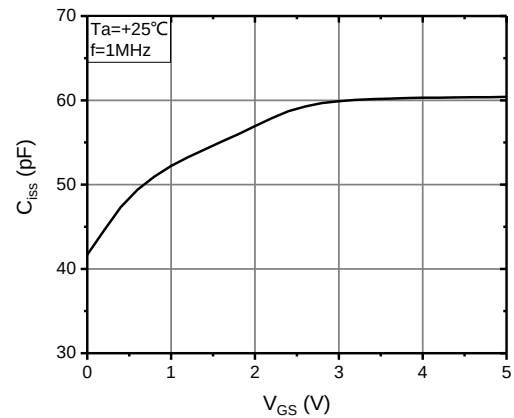
I_{DS} VS. V_{GS}



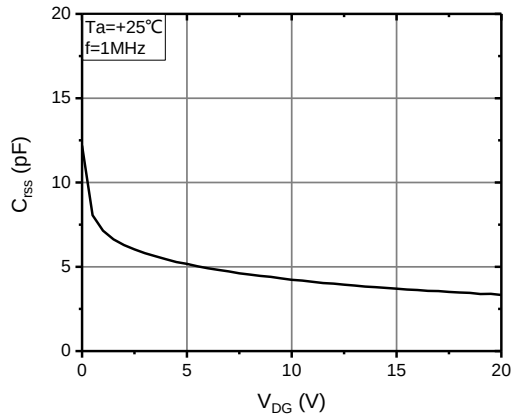
I_{DS} VS. V_{DS}



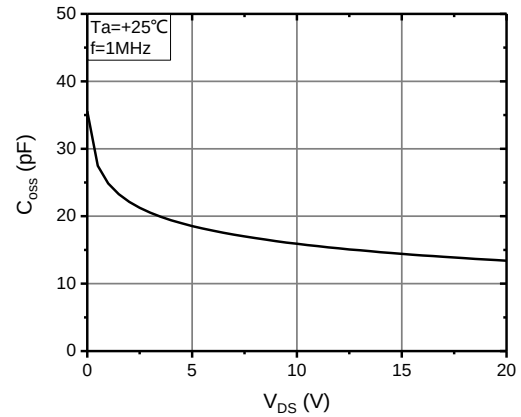
C_{iss} VS. V_{DS}



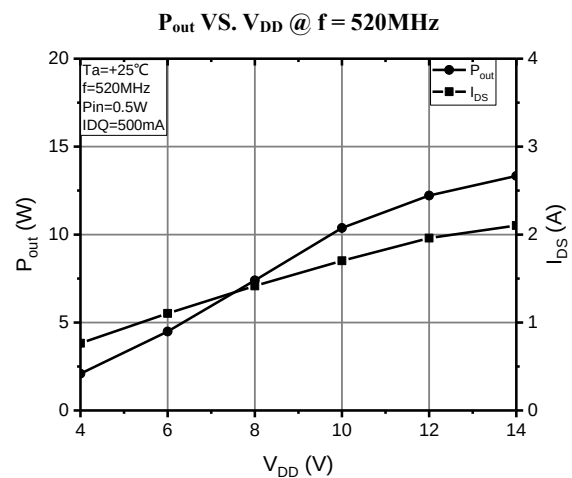
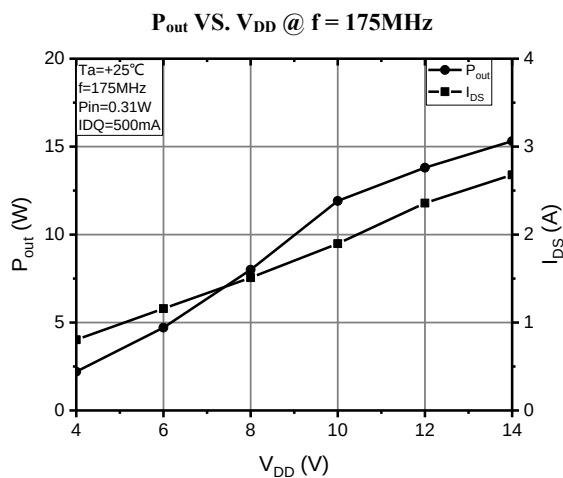
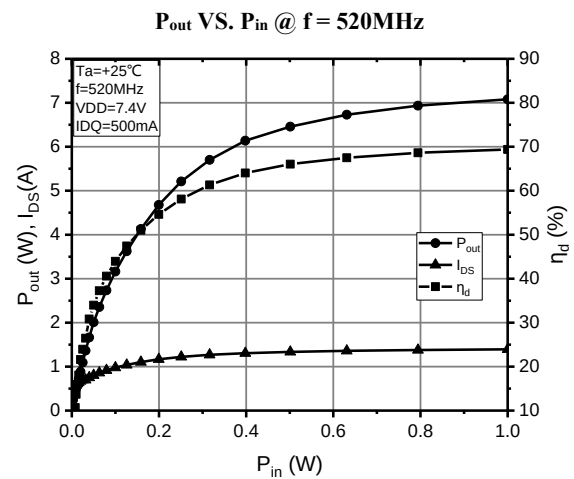
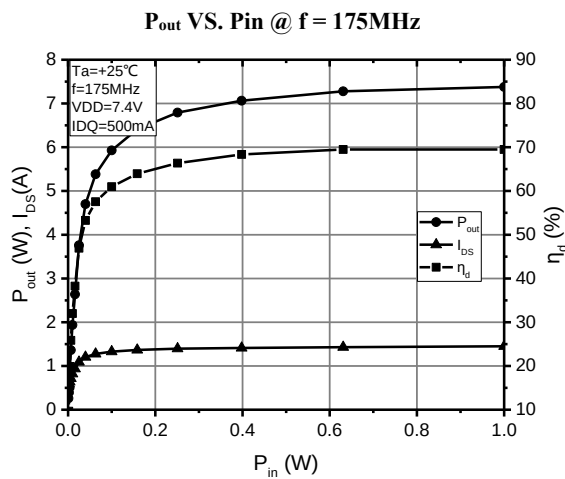
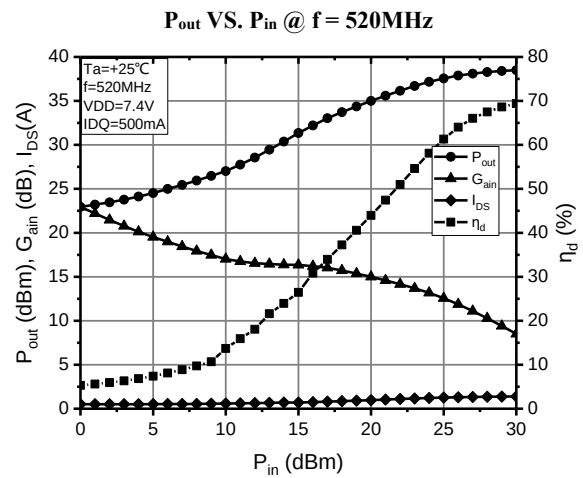
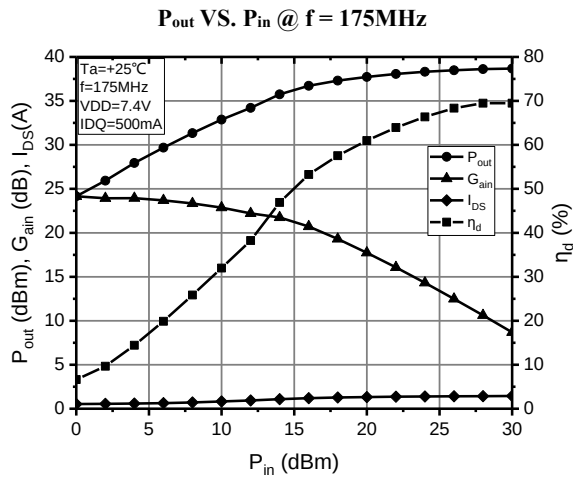
C_{rss} VS. V_{DS}



C_{oss} VS. V_{DS}

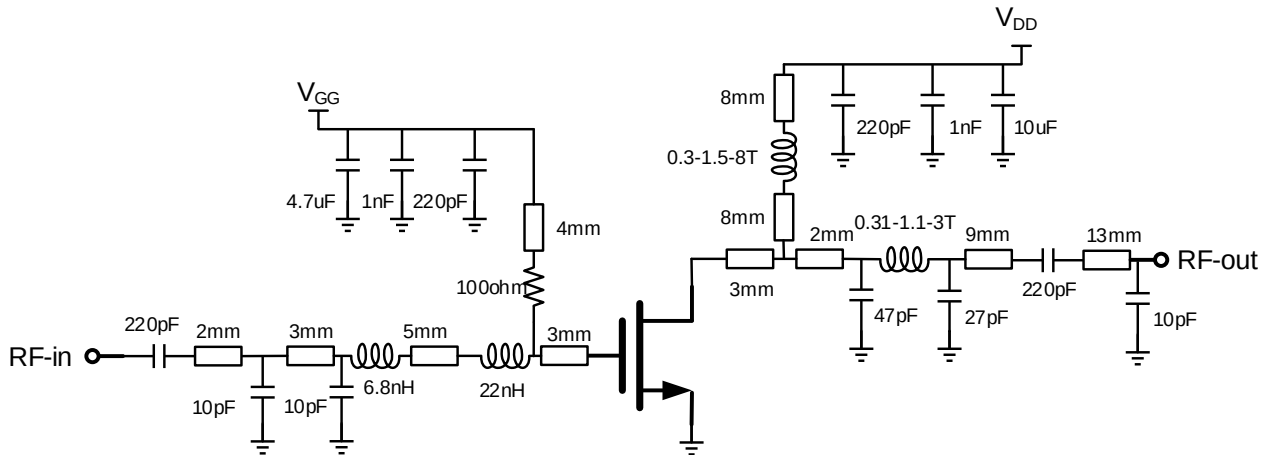


9. Typical Characteristics @ $V_{DD}=7.4V$, VHF/UHF-band



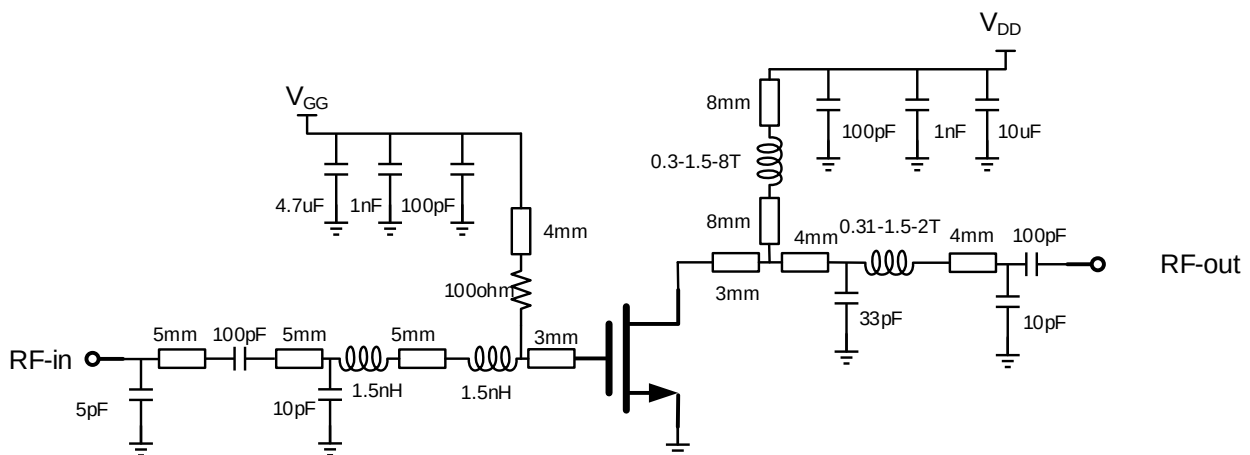
10. Test Circuit @ $V_{DD}=7.4V$, VHF/UHF-band

175MHz @ $V_{DD} = 7.4V$, $I_{DQ} = 500mA$



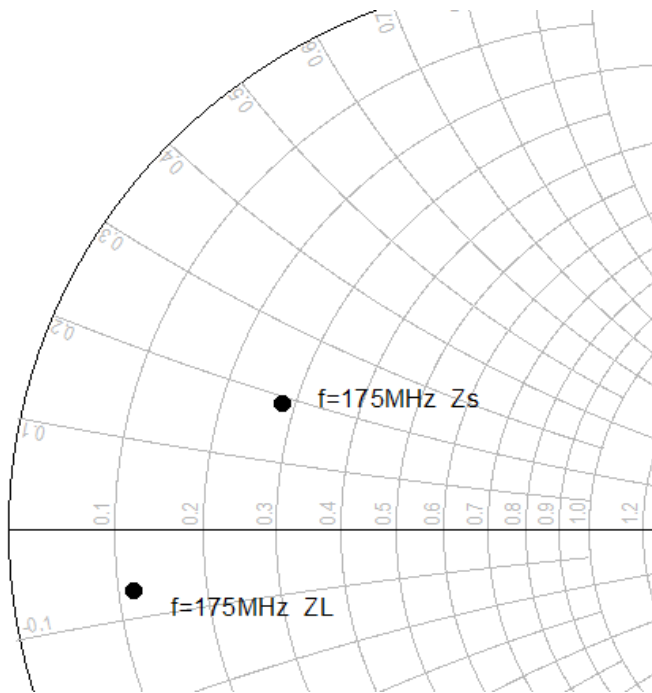
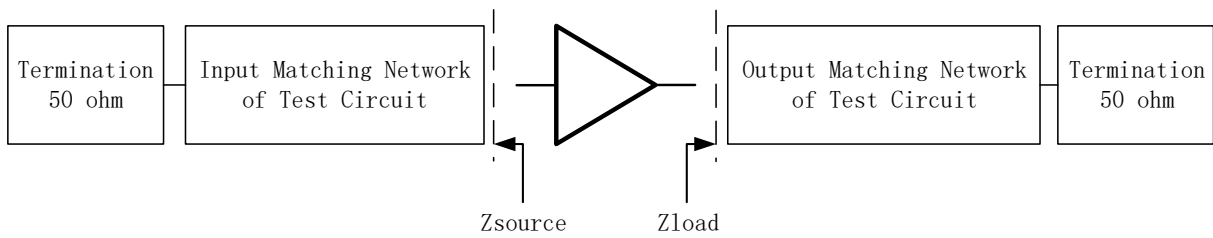
Note: The characteristic impedance of all microstrip lines: 50ohm

520MHz @ $V_{DD} = 7.4V$, $I_{DQ} = 500mA$

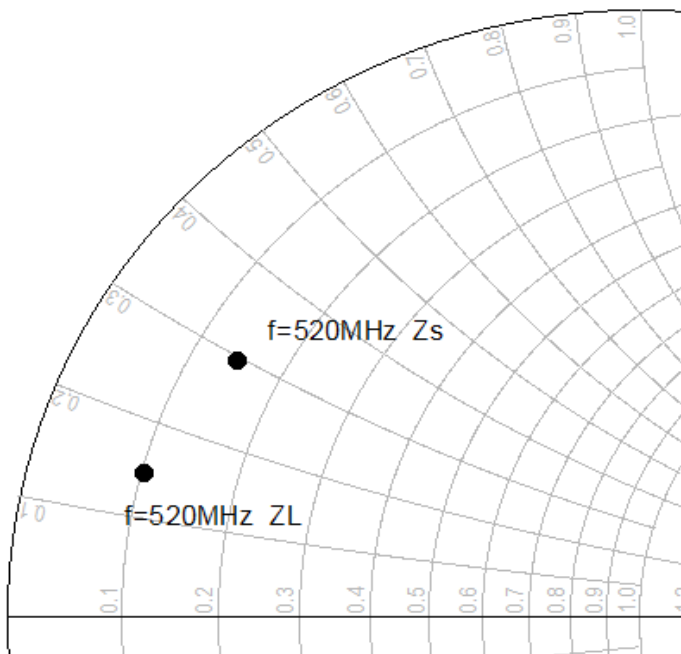


Note: The characteristic impedance of all microstrip lines: 50ohm

11. Input/Output Impedance Characteristics



@ $P_{in}=0.31W$, $V_{DS}=7.4V$, $I_{dq}=500mA$	
Z_s	$13.90 + j 9.38$
Z_L	$5.21 - j 3.41$

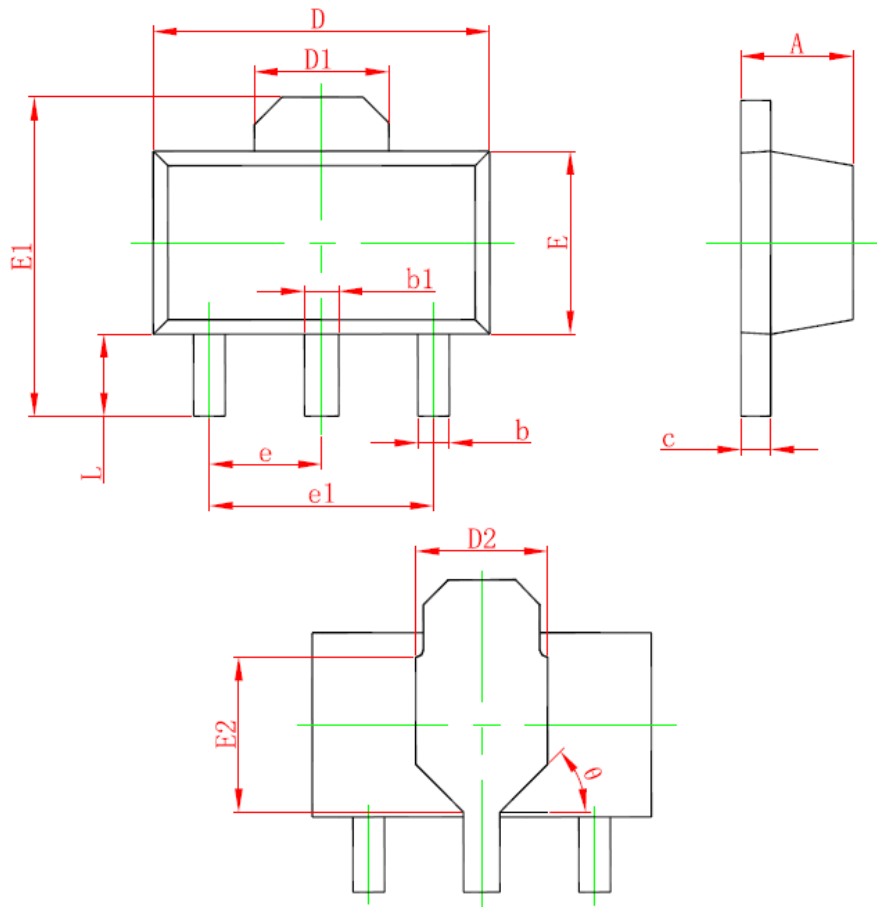


@ $P_{in}=0.5W$, $V_{DS}=7.4V$, $I_{dq}=500mA$	
Z_s	$7.02 + j 15.30$
Z_L	$4.92 + j 7.42$

12.HTL7G06S006P S-Parameter Data ($V_{DD}=7.4V$, $I_{dq}=500mA$)

Freq (MHz)	S11		S21		S12		S22	
	(mag)	(ang)	(mag)	(ang)	(mag)	(ang)	(mag)	(ang)
150	0.99	176.0	7.56	3.5	0.02	-83.3	0.84	15.1
175	0.97	172.4	6.66	-11.2	0.02	-97.2	0.84	-10.1
200	0.96	167.8	5.62	-32.9	0.02	-117.5	0.84	-46.2
250	0.94	161.2	4.44	-68.0	0.02	-151.7	0.84	-106.0
300	0.93	155.3	3.66	-102.2	0.02	175.7	0.84	-165.2
350	0.92	149.6	3.11	-135.8	0.02	144.3	0.83	136.3
400	0.92	144.2	2.70	-169.0	0.02	112.3	0.83	78.0
450	0.92	138.9	2.39	157.8	0.01	80.5	0.83	19.7
500	0.91	133.7	2.16	124.6	0.02	50.4	0.83	-38.6
520	0.91	131.6	2.07	111.2	0.02	37.5	0.84	-62.2
550	0.91	128.6	1.95	91.2	0.02	18.2	0.83	-97.2
600	0.91	123.4	1.77	58.0	0.01	-12.6	0.83	-155.8
650	0.91	118.4	1.62	25.2	0.01	-43.5	0.82	146.2
700	0.90	113.4	1.49	-7.2	0.01	-73.9	0.82	88.8
750	0.90	108.5	1.39	-39.7	0.01	-102.5	0.82	31.7
800	0.90	103.6	1.30	-72.3	0.01	-134.1	0.82	-25.7
850	0.89	98.7	1.23	-105.1	0.01	-164.7	0.82	-83.8
900	0.89	93.9	1.16	-138.2	0.01	162.4	0.81	-142.6
950	0.89	89.1	1.09	-171.1	0.01	131.3	0.81	158.6
1000	0.88	84.5	1.02	156.6	0.01	107.3	0.80	100.7
1050	0.87	80.0	0.97	124.3	0.01	72.7	0.80	43.9
1100	0.87	75.7	0.92	92.5	0.01	42.5	0.79	-12.8
1150	0.87	71.5	0.89	60.4	0.01	12.7	0.80	-70.4
1200	0.87	67.0	0.86	27.8	0.01	-18.1	0.79	-128.5
1250	0.87	62.5	0.83	-4.9	0.01	-49.3	0.79	172.8
1300	0.87	57.9	0.80	-37.5	0.01	-79.2	0.78	114.6
1350	0.87	53.3	0.78	-70.0	0.01	-108.3	0.78	56.6
1400	0.87	48.8	0.75	-102.4	0.01	-140.5	0.79	-0.8
1450	0.86	44.3	0.74	-135.0	0.01	-172.2	0.79	-58.6
1500	0.86	39.8	0.72	-167.7	0.01	158.4	0.78	-117.6

13.Recommended PCB Pad Layout



Symbol	Dimesions in Milimeters		Dimesions in Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF.		0.061 REF.	
D2	1.750 REF.		0.069 REF.	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
E2	1.900 REF.		0.075 REF.	
e	1.500 TYP.		0.060 TYP.	
e1	3.000 TYP.		0.118 TYP.	
L	0.900	1.200	0.035	0.047
θ	45°		45°	

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